KHULNA
UNIVERSITY OF
ENGINEERING AND
TECHNOLOGY

Akash Biswas
Lecturer, EEE
KUET

Department of Electrical and Electronic Engineering

Course Code : EE 4121

VLSI

Introduction to VLSI Technology

2.15 CMOS Pass Gate

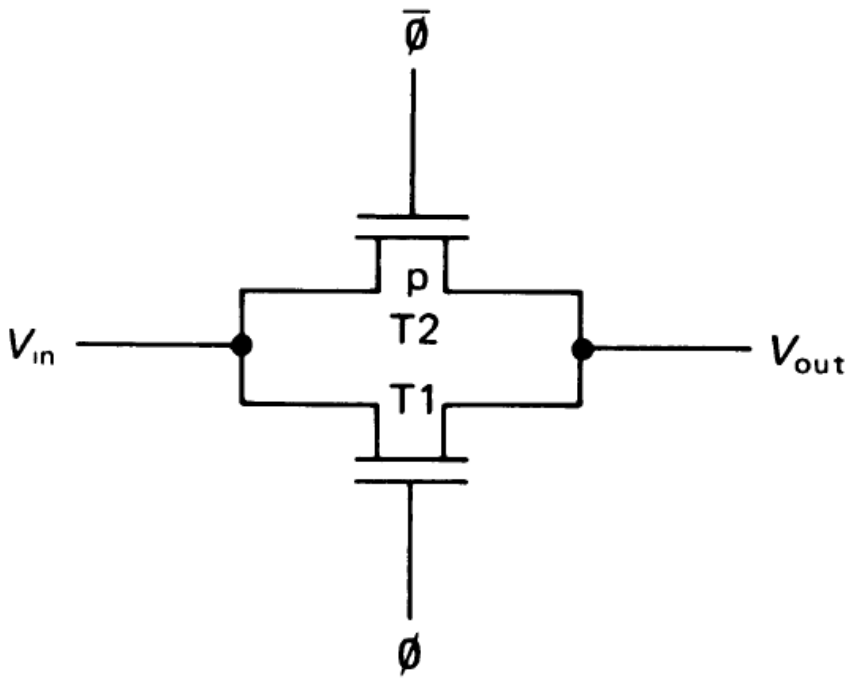


Figure 2.26 CMOS pass gate

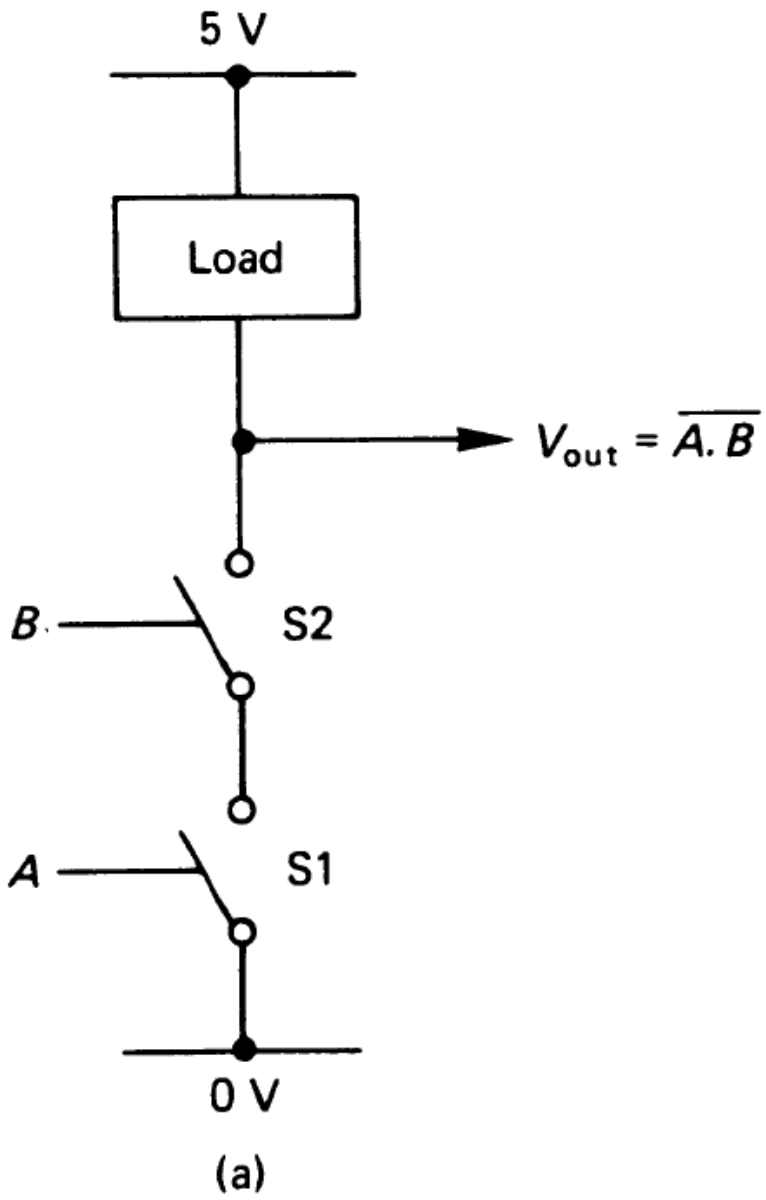
When ϕ is taken high, both devices turn on. Again there are four cases to consider.

(a) $V_{in} = 0$ V, initial $V_{out} = 0$ V
The drain and source potentials of each device are equal, so no current flows. V_{out} remains at 0 V.

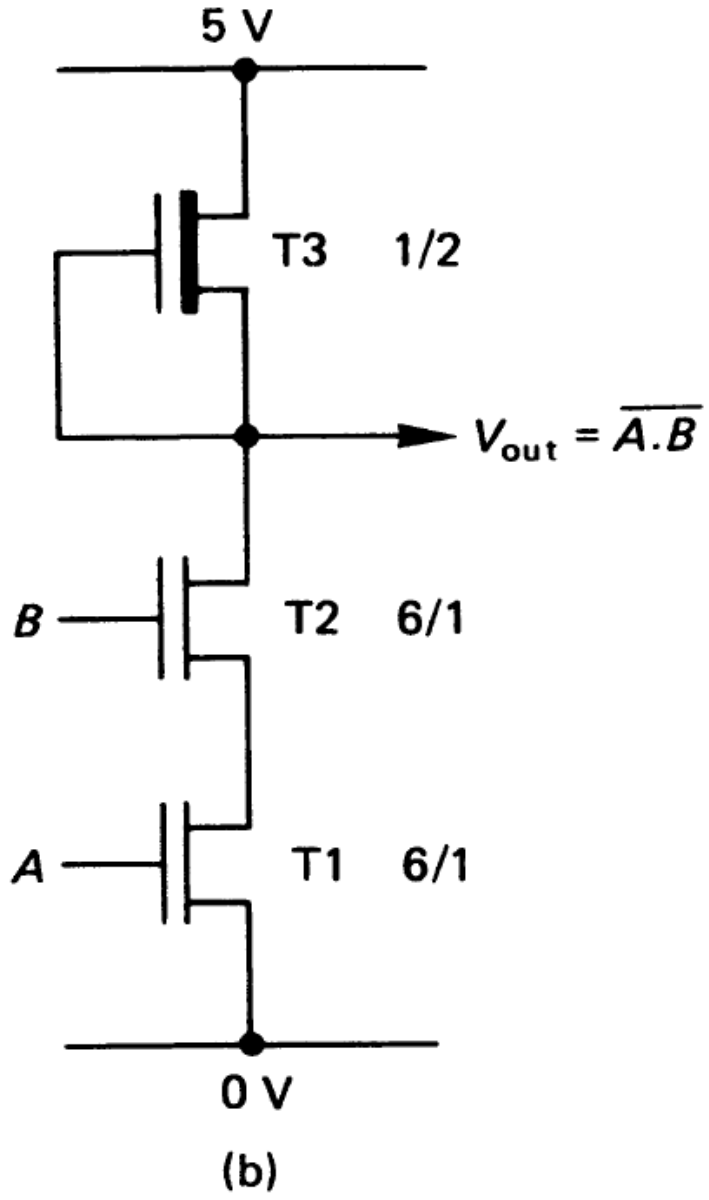
(b) $V_{in} = V_p$, initial $V_{out} = V_p$
Again the drain and source potentials of T1 and T2 are equal, so no current flows and V_{out} remains at V_p.

(c) $V_{in} = V_p$, initial $V_{out} = 0$ V
Here V_{in} acts as the drain of T1 and the source of T2. Current flows from V_{in} to V_{out}, causing V_{out} to rise as the capacitance C_{out} on the output charges up.

(d) $V_{in} = 0$ V, initial $V_{out} = V_p$
V_{in} acts as the source of T1 and the drain of T2. Current flows from V_{out} to V_{in}, causing V_{out} to fall as C_{out} discharges.


5 V
Load
$V_{out} = \overline{A.B}$
B
S2
A
S1
0 V

(a)


5 V
T3 1/2
$V_{out} = \overline{A.B}$
B
T2 6/1
A
T1 6/1
0 V

(b)

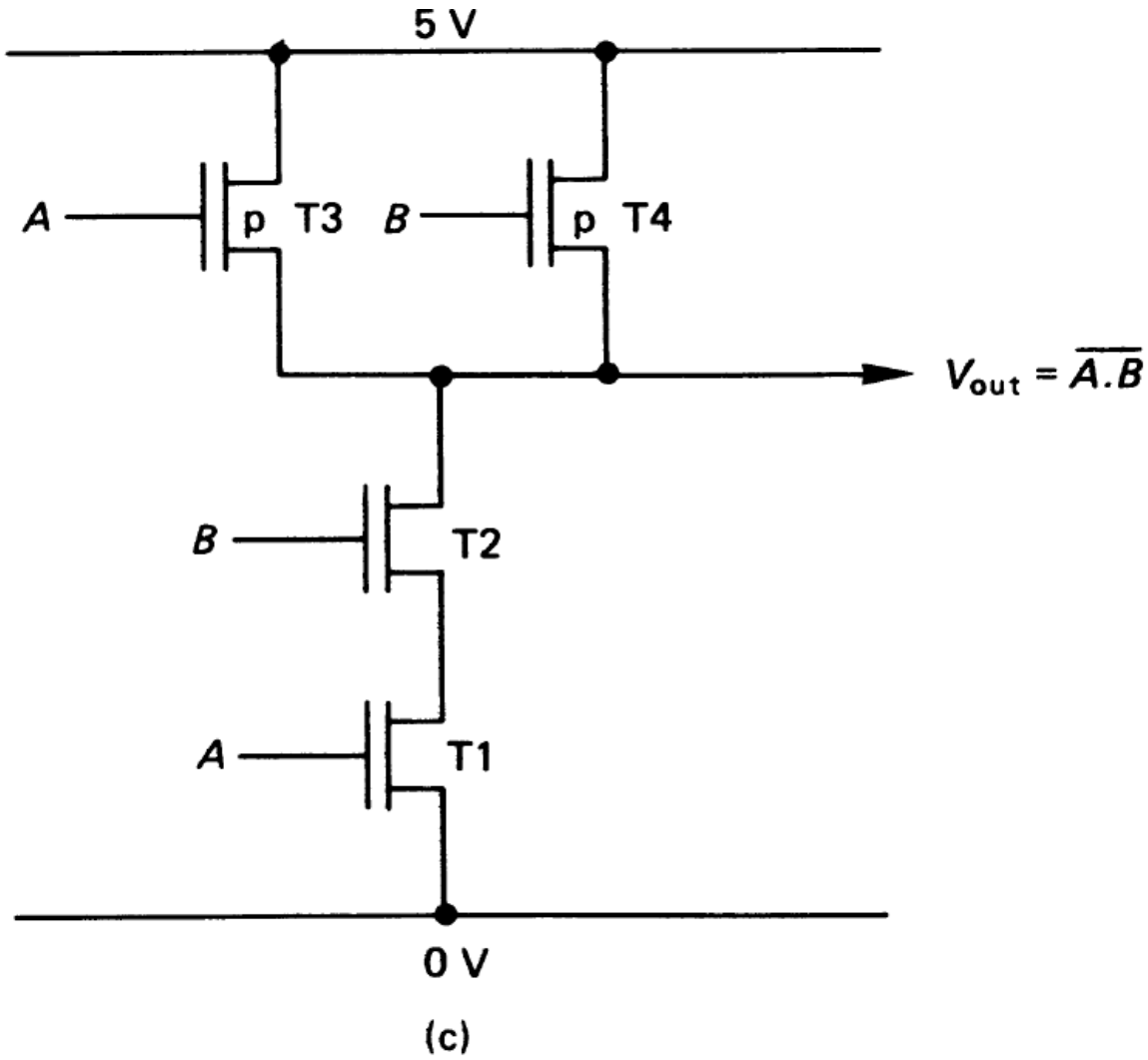
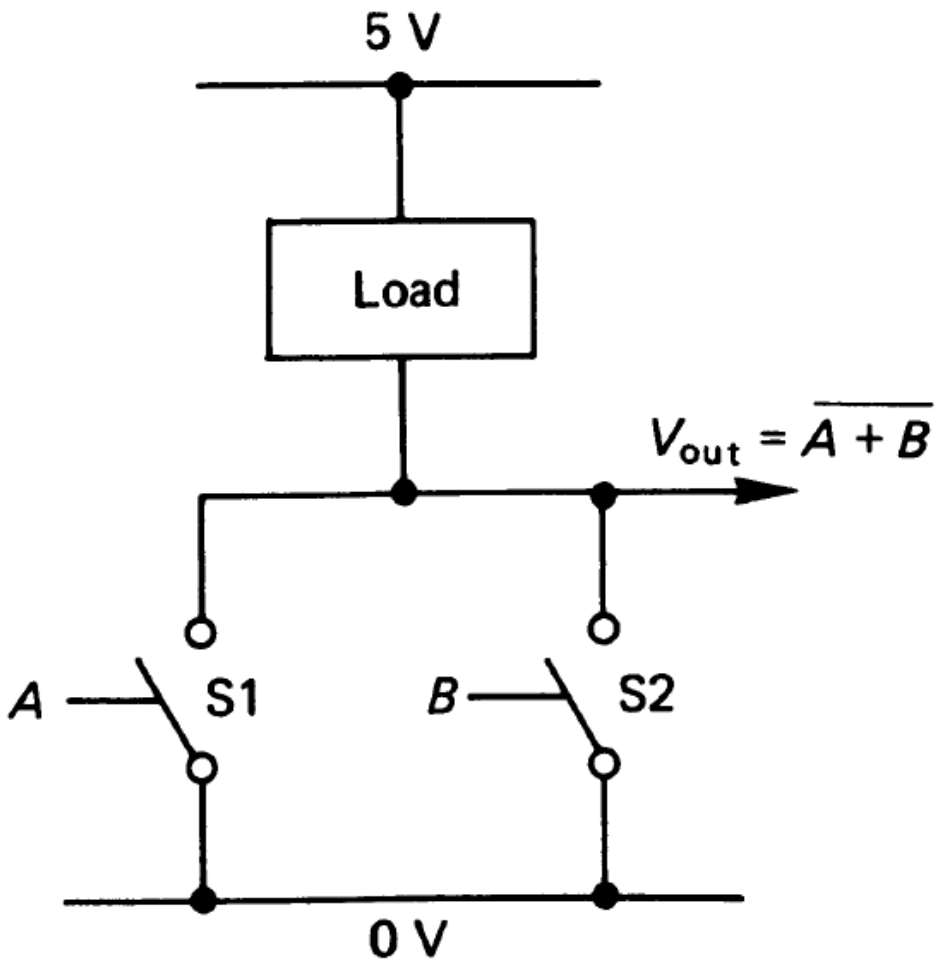
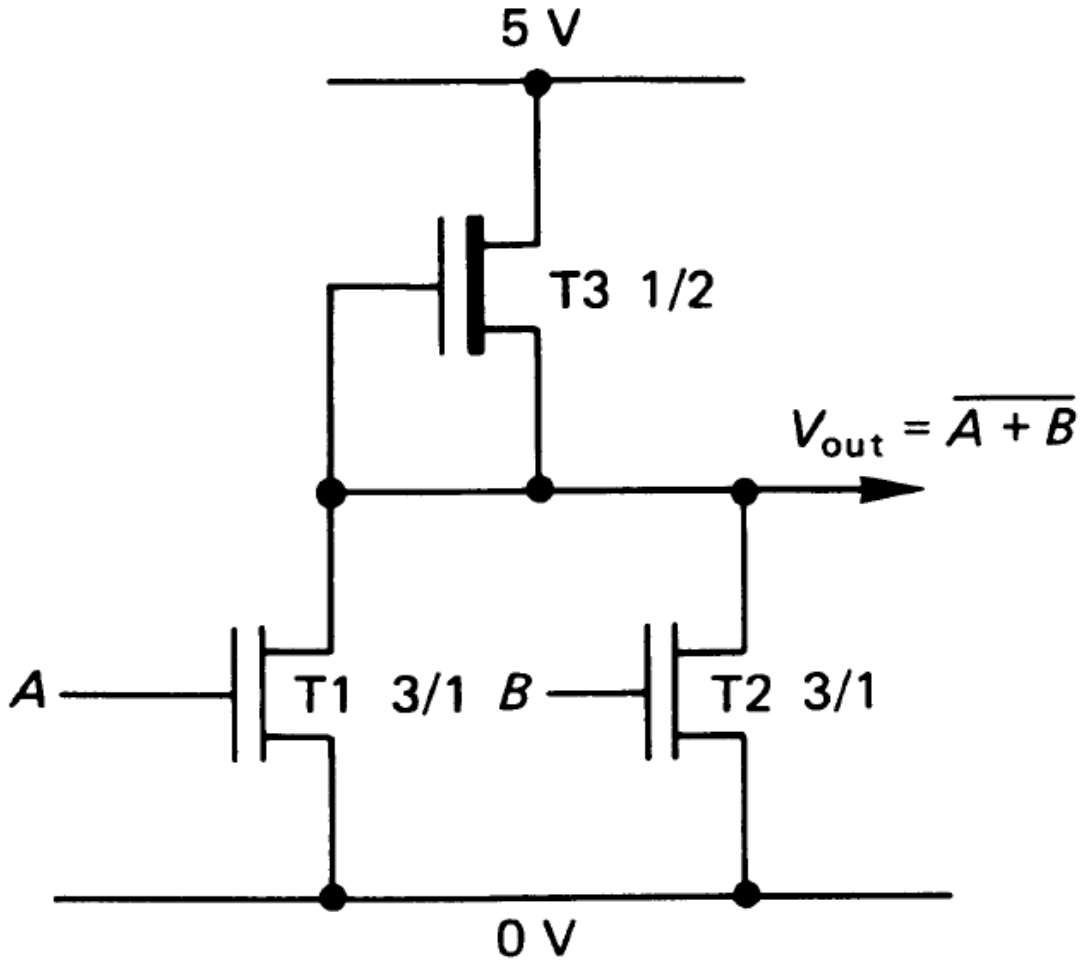


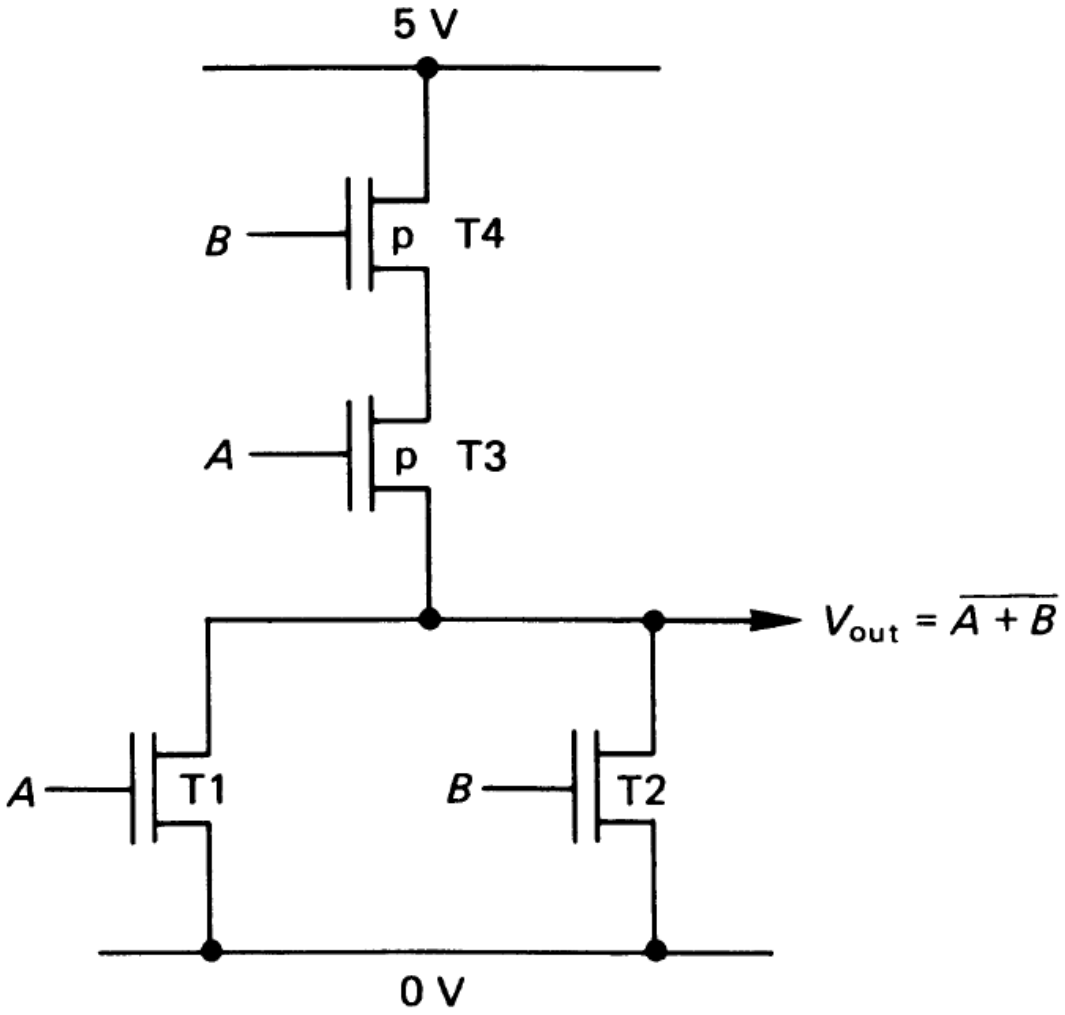
Figure 4.1 A two-input nand gate: (a) switches, (b) NMOS technology,


5 V
Load
$V_{out} = \overline{A + B}$
A
S1
B
S2
0 V

(a)


5 V
T3 1/2
$V_{out} = \overline{A + B}$
A
T1 3/1
B
T2 3/1
0 V

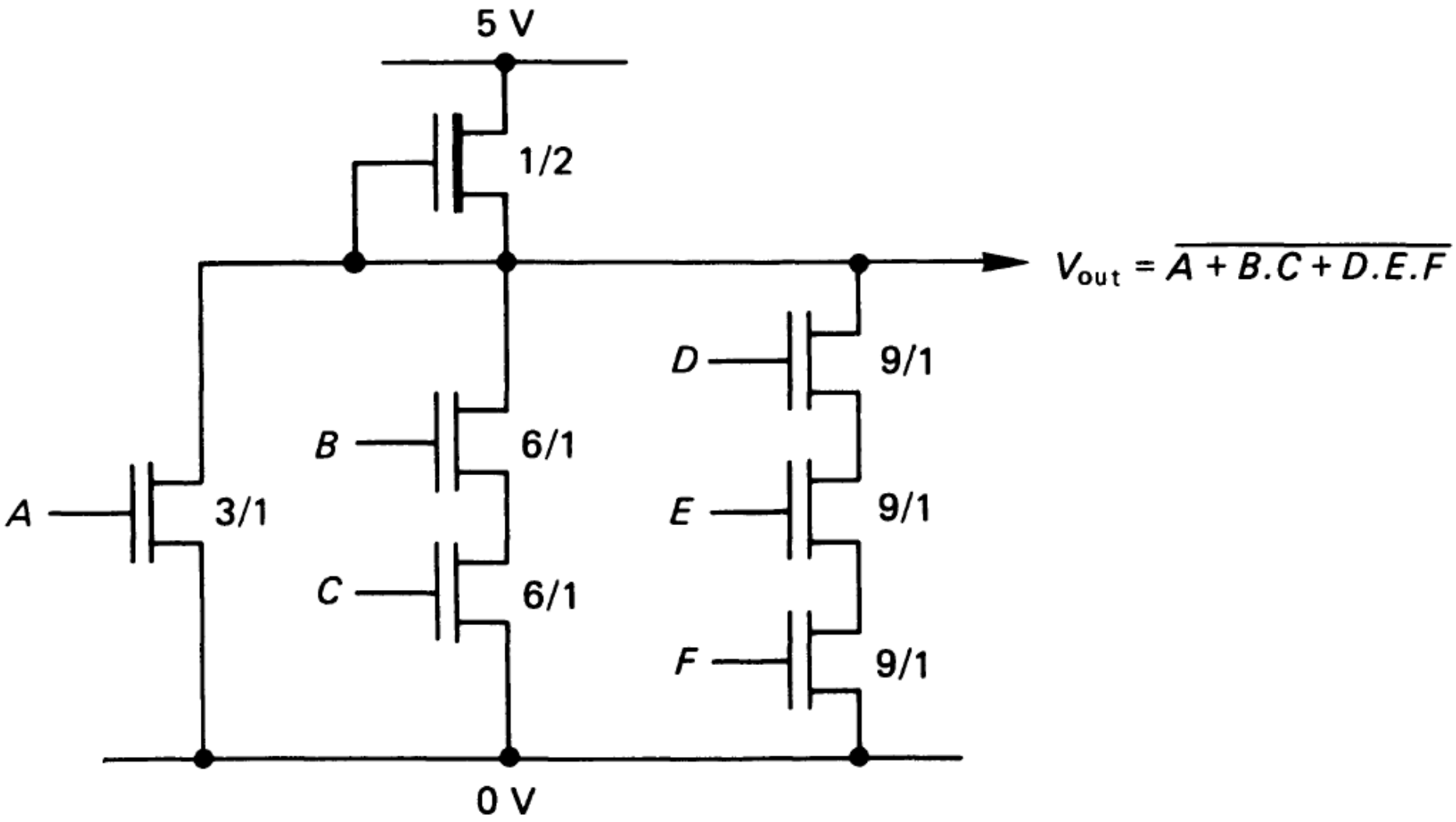
(b)

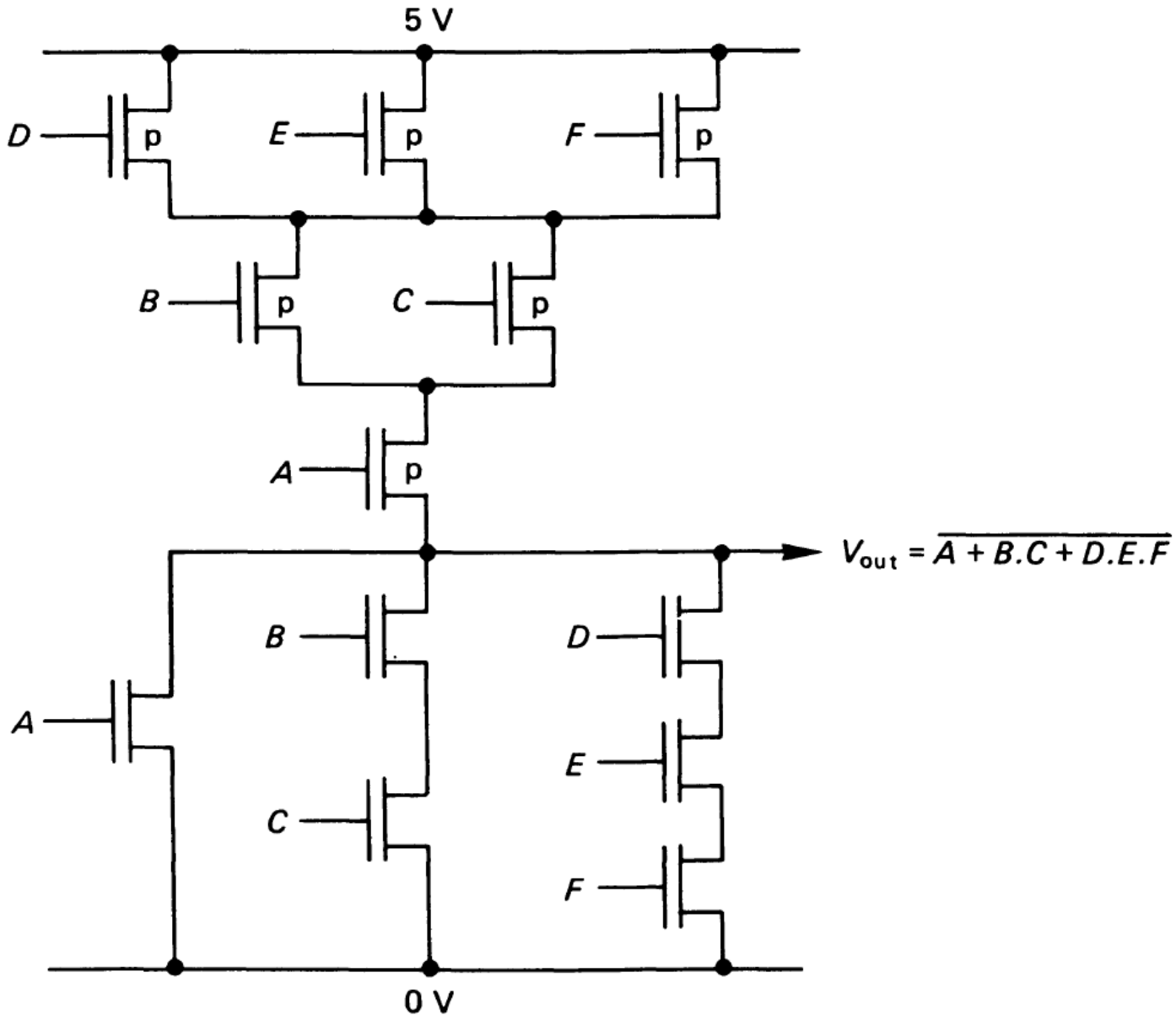


(c)

Figure 4.2 A two-input nor gate: (a) switches, (b) NMOS technology (c) CMOS circuit

$$V_{out} = \overline{A + B.C + D.E.F}$$




5 V
D
p
E
p
F
p
B
p
C
p
A
p
$V_{out} = \overline{A + B.C + D.E.F}$
B
D
A
E
C
F
0 V

Euler path:

***1. With identical ordering of input labels find a path for both PUN and PDN.

***2. Traverse each edge (branch) of the graph exactly once, path must be uninterrupted. [node on vertex can be traversed more than once but not branches]

3. Traversing can be clockwise or anticlockwise for both PUN and PDN network, it does not really matter, but ordering must be same.
if no common ordering path -no Euler path

4. A complex CMOS logic without Euler path can be constructed to Stick-diagram layout but there will be active area break. (eg- y=AB+BD+EF)
Euler path method is the simple method for finding optimum gate ordering.

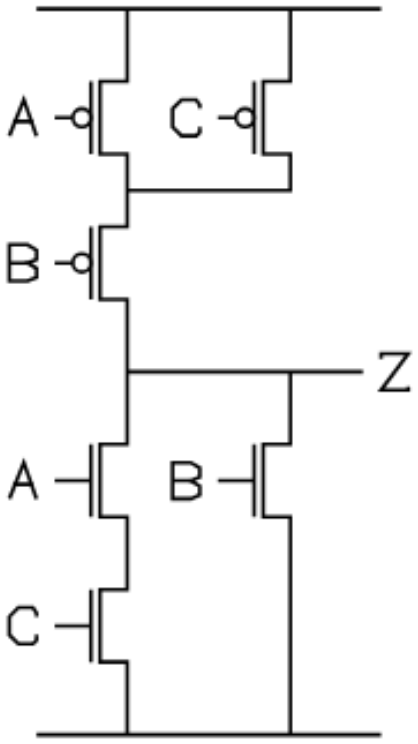
* CMOS logic or diagram having euler path can also be constructed to stick-diagram layout without maintaining euler path, but then Layout will not be optimum.(will have break)

SOURCE: https://www.southampton.ac.uk

Finding an Euler Path

$Z = \overline{(A \bullet C) + B}$

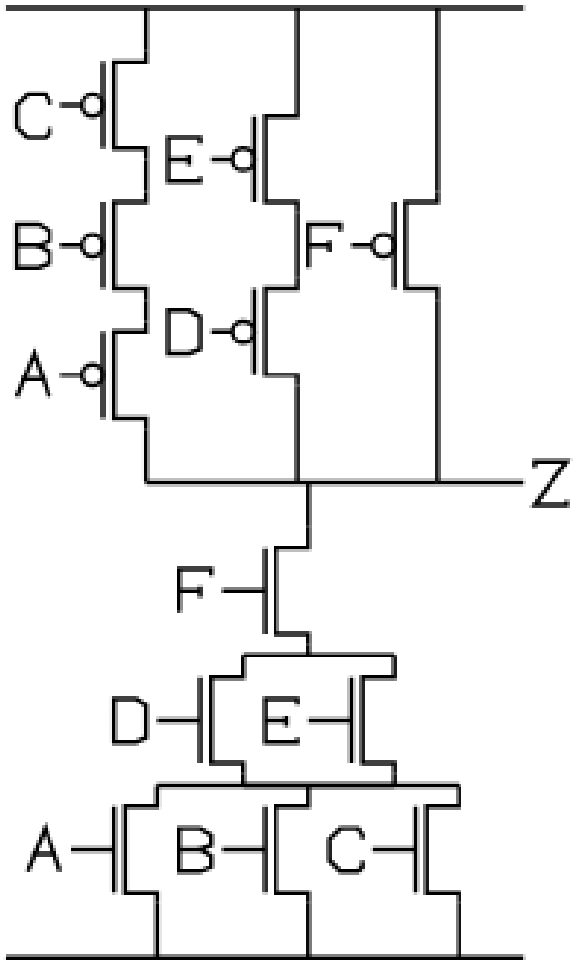




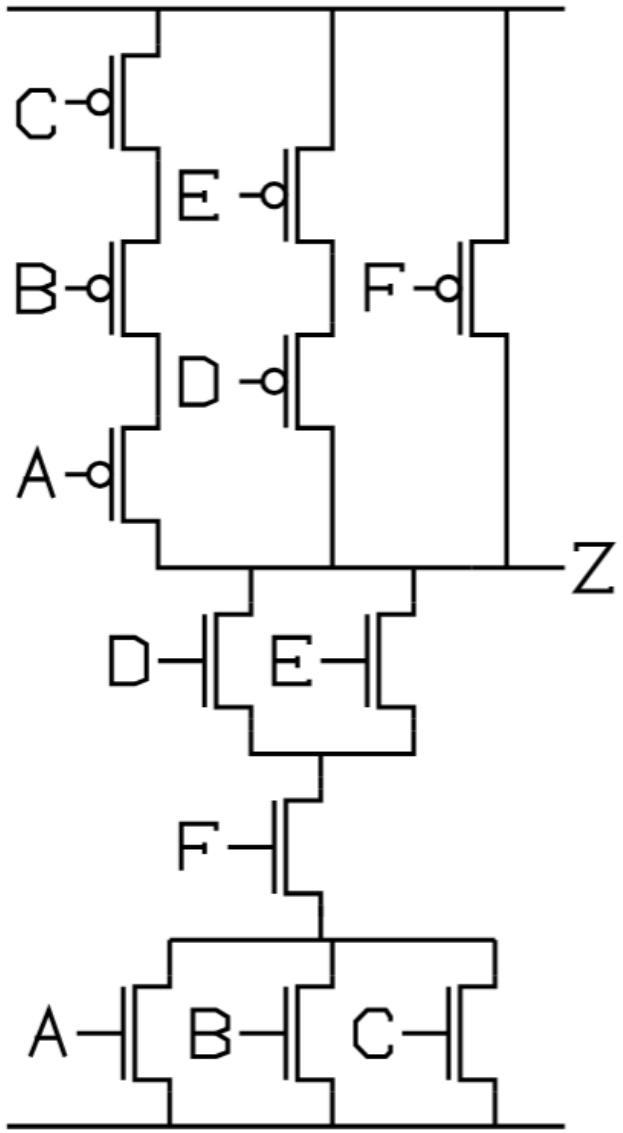
A–C–B
B–A–C
B–C–A
C–A–B ⇐

A–B–C
A–C–B
B–A–C
B–C–A
C–A–B ⇐
C–B–A

$$Z = \overline{(A+B+C) \bullet (D+E) \bullet F}$$

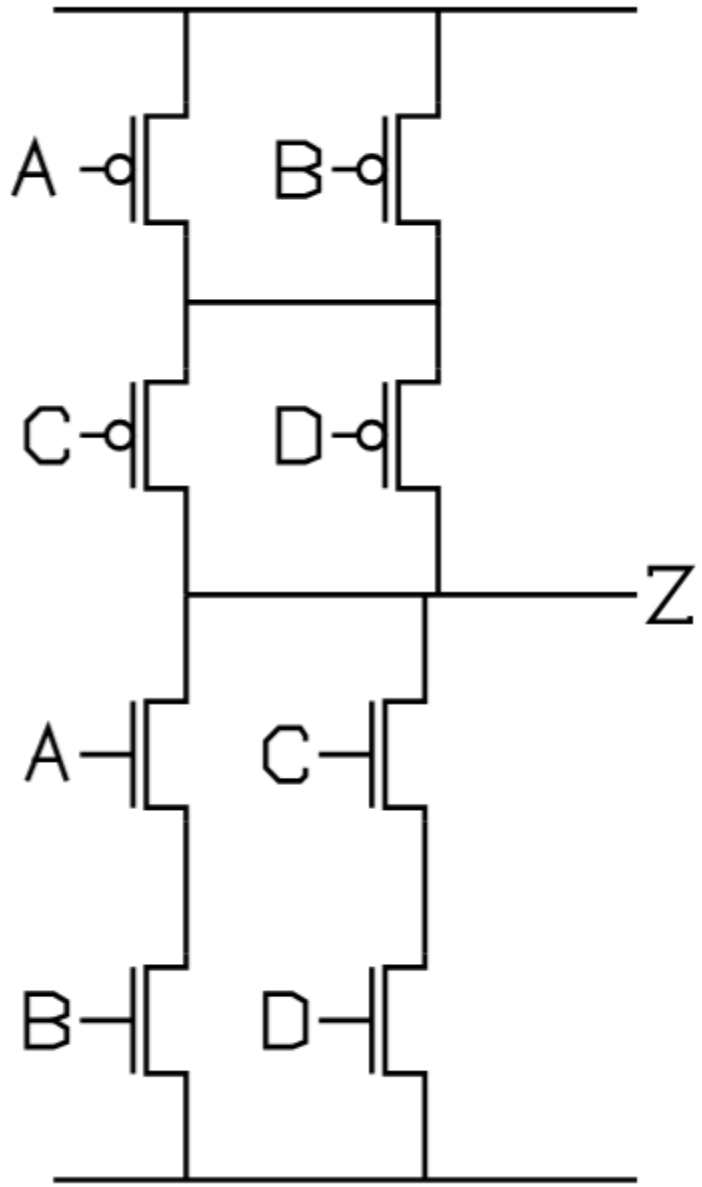


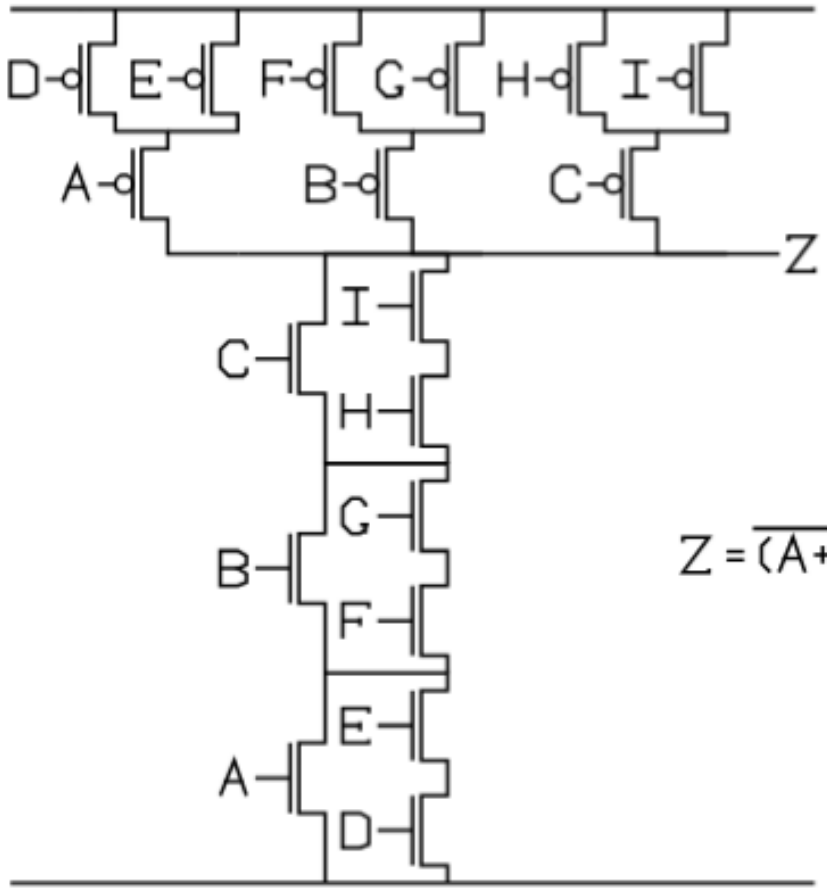
Does Not maintain Euler path



$$Z = \overline{(A+B+C) \bullet (D+E) \bullet F}$$

$$Z = \overline{(A \bullet B) + (C \bullet D)}$$





$$Z = \overline{(A+(D \bullet E)) \bullet (B+(F \bullet G)) \bullet (C+(H \bullet I))}$$

- No possible path through p-transistors.
- No re-arrangement will create a solution!